Branch Coverage Report

Project: Task Manager Application

Version: 2.0 | Section: J | Batch: 12 | Date: 05-11-2026 | Build under test: `v1.0.1` (Sprint 3)

Authors

SRN	Name
PES2UG24CS561	Tharunyah Sudhakar
PES2UG24CS594	Vishnu Narayanan Ajay
PES2UG24CS597	Vismay Chandra Dev
PES2UG24CS606	Yajat Mathur

Revision History

Version	Date	Author	Change Summary
1.0	04-11-2026	Student Team	Branch coverage for tag v1.0.
2.0	05-11-2026	Student Team	Branch coverage for v1.0.1.

Approvals

Role	Name	Signature / Date
QA Lead	Tharunyah Sudhakar	
Course Coordinator		

1. Result

Test set	Branches	Taken	Not taken	Branch coverage
v1.0 code, tests as tagged	196	182	14	92.9%
v1.0 code, with Part 3 tests	196	196	0	100.0%
v1.0.1 code, all tests	220	220	0	100.0%

Measured with: coverage.py 7.17.2, over the full unit test suite (155 tests, 155 passed). Line numbers in the gap tables refer to the v1.0 source.

A branch is one possible path out of a decision: the true or false side of an `if`, or entering or leaving a `for` loop. Conditions inside a single line, such as `x if y else z`, are not counted separately, as in coverage.py's line-level view.

2. Branch coverage by module

Module	Branches	Taken	Not taken	Coverage	Branches not taken (from line to line)
app/__init__.py	10	10	0	100.0%	None
app/access.py	4	4	0	100.0%	None
app/admin.py	20	20	0	100.0%	None
app/auth.py	12	12	0	100.0%	None
app/config.py	0	0	0	n/a	None
app/dashboard.py	0	0	0	n/a	None

Module	Branches	Taken	Not taken	Coverage	Branches not taken (from line to line)
app/db.py	4	4	0	100.0%	None
app/errors.py	4	4	0	100.0%	None
app/notifications.py	28	28	0	100.0%	None
app/scheduling.py	28	28	0	100.0%	None
app/search.py	14	14	0	100.0%	None
app/security.py	18	18	0	100.0%	None
app/status.py	0	0	0	n/a	None
app/tasks.py	24	24	0	100.0%	None
app/utils.py	6	6	0	100.0%	None
app/validators.py	48	48	0	100.0%	None

3. Branches not taken by the v1.0 tests and closed in Part 3

Module	Branch	Decision at the start of the branch
app/admin.py	108 to 111	`if request.args.get("type"):`
app/errors.py	69 to 72	`if request.path.startswith("/api/"):`
app/notifications.py	57 to 58	`if not current_app.config["EMAIL_NOTIFICATIONS_ENABLED"]:`
app/notifications.py	57 to 60	`if not current_app.config["EMAIL_NOTIFICATIONS_ENABLED"]:`
app/notifications.py	104 to 106	`if channel == "email" and not email_notifier.send(`
app/notifications.py	125 to 123	`if count:`
app/notifications.py	125 to 126	`if count:`
app/security.py	84 to 87	`if db is not None:`
app/tasks.py	179 to 180	`if not changes and new_categories is None:`
app/utils.py	55 to 57	`if dt.tzinfo is not None: # convert offsets such as +05:30 to UTC`
app/validators.py	42 to 43	`if len(name) > MAX_NAME:`
app/validators.py	77 to 78	`if not isinstance(value, str):`
app/validators.py	79 to 80	`if len(value) > MAX_DESCRIPTION:`
app/validators.py	149 to 150	`if not name:`

What these figures do not show

- **Scope:** the Python package `app/`, as in CI. `run.py` and the front-end JavaScript in `app/static/` are not measured here; the front-end is exercised by the browser checks in the System Validation Report.
- **Covered does not mean correct.** The Part 3 tests ran every statement of v1.0 while defects DEF-01 to DEF-05 were still in that code: coverage shows that code ran, not that every input is handled. Sprint 3 therefore added route-wide boundary and fuzz tests.